

# CarvAcoustic

## Planning Packet

Report 1: Software Product Execution and Staging Plan

Report 2: Technical Build Specification for Phase 1

Plus compact LLM handoff packet

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>Product</b>             | CarvAcoustic                                                            |
| <b>Domain</b>              | carvacoustic.com                                                        |
| <b>Current build focus</b> | Decorative wall art, cabinet/front panels,<br>architectural face panels |
| <b>Downstream CAM</b>      | Vectric remains downstream in phase 1                                   |

Prepared as a compact planning and engineering handoff package for AI-assisted implementation.

# What's in this package

This document is the human-readable share version. The companion ZIP includes the two full reports in Markdown plus a five-file LLM handoff packet that is easier to feed into coding agents with limited context windows.

| Included file            | Purpose                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| Report 1                 | Software product execution, staging, stage gates, workflow proof, website role, acoustic roadmap framing |
| Report 2                 | Technical build spec for phase 1, including schemas, geometry system, API, UX, export and QA             |
| 00-governing-decisions   | Canonical scope, no-build rules, gate criteria, coding constraints                                       |
| 01-schema-contracts      | Entities, canonical config, versioning, storage, reserved acoustic fields                                |
| 02-geometry-spec         | Approved modes, pattern families, pipeline, validation, layout, export layers                            |
| 03-api-and-ux            | Route map, UI layout, endpoint contracts, Vectric handoff model                                          |
| 04-build-order-and-tests | Implementation sequence, milestones, golden samples, prompt pack, QA                                     |

## Recommended use order

- Read Report 1 first to lock phase boundaries and stage gates.
- Use Report 2 to build phase 1 without reopening product-scope questions.
- Feed the compact packet files to coding agents in the order 00 → 04.

# Report 1 — Software Product Execution and Staging Plan

## Executive summary

- CarvAcoustic should begin as a web-first software tool with an authenticated app shell that generates decorative panel geometry, validates manufacturability, nests parts, and exports Vectric-ready files.
- Phase 1 proves decorative workflows first. Acoustic features are staged later because they require different inputs, validation logic, and installation constraints.

- The website is part of the product from day one: landing page, app shell, dashboard, docs, and export/download surface.

## **Product intent**

- Phase 1 is a design-to-export system for decorative wall art, decorative cabinet/front panels, and decorative architectural face panels.
- Phase 1 is not a full CAD system, not a CAM replacement, not a G-code generator, and not the manufacturing or marketing system.

## **Scope boundaries**

- Included: software planning, website planning, phased workflow planning, stage gates, founder proof criteria, AI-agent guidance, acoustic roadmap framing, and future intake planning at a high level.
- Excluded: manufacturing operations, packaging, shipping, pricing, full acoustic implementation, and marketing strategy.

## **Product phases and gates**

- Phase 0: define goals, scope, architecture boundaries, sample projects, and coding rules.
- Phase 1: prove the decorative workflow end to end.
- Phase 2: harden reliability, presets, UX, validation, and exports.
- Phase 3: prepare acoustic intake and schema extensions without implementing acoustic generation.
- Phase 4: implement acoustic features after decorative workflow is stable.

## **Gate criteria**

- Gate 1 requires one valid wall-art project, one valid cabinet/front-panel project, clean Vectric import, useful nesting, save/load/versioning, and validation that catches common failures.
- Gate 2 requires repeated founder usage with minimal and predictable Vectric cleanup before acoustics begin.

## **Website role**

- Minimum phase-1 scope: home page, auth, dashboard, create-project flow, design workspace, validation/export surface, docs/help, and founder preset controls.
- Customer intake, vendor portals, and acoustic measurement upload flows are reserved for later phases.

## Acoustic expansion framing

- Acoustic work stays in separate future modes. Future intake should support simple self-service room data and more advanced vendor-assisted measurements.

## Report 2 — Technical Build Specification for Phase 1

### Canonical technical decisions

- Use the product name CarvAcoustic and domain carvacoustic.com.
- Keep Vectric downstream in phase 1.
- Prioritize geometry generation, manufacturability validation, export quality, project persistence, and workflow usability.
- Do not build acoustic generation, cloud suspension, G-code, manufacturing operations, or marketing automation in phase 1.

### Phase 1 must build

- Web app shell with auth, dashboard, project list, create/edit flow.
- Design modes: wall\_art, cabinet\_front\_panel, architectural\_face\_panel.
- Boundary inputs: rectangle, rounded rectangle, imported SVG.
- Pattern families: wave\_field, contour\_bands, slat\_rib.
- Processing: geometry generation, validation, sheet layout, export bundle.
- Persistence: save/load, immutable project versions, presets.
- Exports: DXF, SVG, PDF reference, JSON manifest.

### Architecture

- Next.js App Router on Vercel for the web app and BFF layer.
- Supabase for Auth, Postgres, and Storage.
- Separate Python FastAPI geometry service using Shapely and ezdxf.
- Synchronous generate/validate/layout/export in phase 1 unless measured latency forces async jobs.

### Repository and domain model

- Monorepo with separate web app and geometry service, plus shared schema/type packages and in-repo docs.
- Core entities: project, project\_version, boundary, pattern, material\_preset, tool\_preset, validation\_report, part, sheet\_layout, export\_bundle, asset, reserved\_acoustic\_intake.

## Geometry engine

- Pure transformation pipeline: config → normalized boundary → generated pattern → validated parts → layout → export artifacts.
- Approved phase-1 pattern families only: wave\_field, contour\_bands, slat\_rib.
- Boundary normalization must reject self-intersections, zero-area shapes, and disconnected outer loops.

## Validation and layout

- Errors block export. Warnings allow export with risk notes. Info is advisory.
- Phase 1 uses usable deterministic sheet layout, not optimal polygon nesting.

## Exports and Vectric handoff

- Per-sheet DXF and SVG, reference PDF, manifest JSON, and project-config JSON in a ZIP bundle.
- Layer structure must remain consistent and Vectric-friendly.
- Founder still handles toolpath assignment, tabs/holds, simulation, and post-processing in Vectric.

## Reserved acoustic architecture

- Keep separate future acoustic modes and reserved config fields but do not implement acoustic logic in phase 1.

## QA and build order

- Build schemas first, then app shell, geometry service, pattern families, validation, layout, export bundle, and acceptance tests.
- Use two golden sample projects: wall\_art\_wave\_panel and cabinet\_front\_contour\_panel.

## Compact LLM packet guidance

The files below are the preferred context slices for coding agents. They are intentionally short and modular so the agent can load only the needed packet instead of the entire planning history.

- **00-governing-decisions.md**: Scope, gates, non-goals, coding rules
- **01-schema-contracts.md**: Canonical config, entities, versioning, storage
- **02-geometry-spec.md**: Modes, pattern families, pipeline, validators, layout, exports
- **03-api-and-ux.md**: Route map, UI structure, API actions, Vectric handoff
- **04-build-order-and-tests.md**: Milestones, golden samples, prompt pack, acceptance checks

## Reference notes

The planning and stack choices in this packet are anchored to official documentation for Vetric, Next.js, Supabase, Shapely, ezdxf, and FastAPI. The ZIP package includes a SOURCES.md file with the exact links used.

End of share document.